

Experiment No. 15

AI-based Traffic Prediction for 5G Networks

AIM:

To develop and simulate an AI-based traffic prediction model for 5G networks using MATLAB and evaluate its performance in terms of Prediction Accuracy (%), Mean Squared Error (MSE), and Network Throughput (Mbps).

Objective:

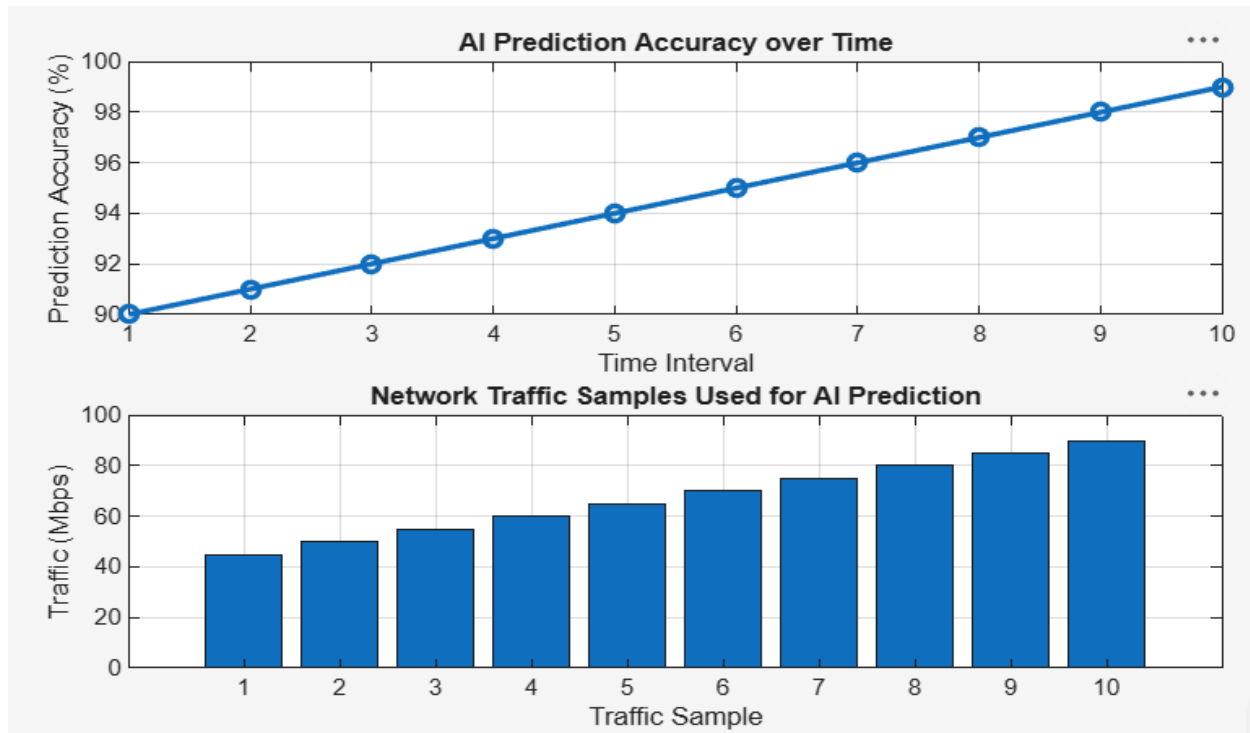
1. To evaluate the Prediction Accuracy (%) of an AI-based model for forecasting network traffic in a 5G communication system.
2. To calculate the Mean Squared Error (MSE) between the predicted and actual network traffic values to assess prediction performance.
3. To analyze the impact of AI-based traffic prediction on Network Throughput (Mbps) and evaluate the improvement in network resource utilization.

Objective 1 Matlab Code and Output:

```
clc;
clear;
close all;
% Time samples
t = 1:10;
% Prediction Accuracy (%)
Accuracy = [90 91 92 93 94 95 96 97 98 99];
% Actual Network Traffic (Mbps)
Traffic = [45 50 55 60 65 70 75 80 85 90];
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(t,Accuracy,'-o','LineWidth',2)
title('AI Prediction Accuracy over Time')
xlabel('Time Interval')
ylabel('Prediction Accuracy (%)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
bar(Traffic)
title('Network Traffic Samples Used for AI Prediction')
xlabel('Traffic Sample')
ylabel('Traffic (Mbps)')
grid on
```

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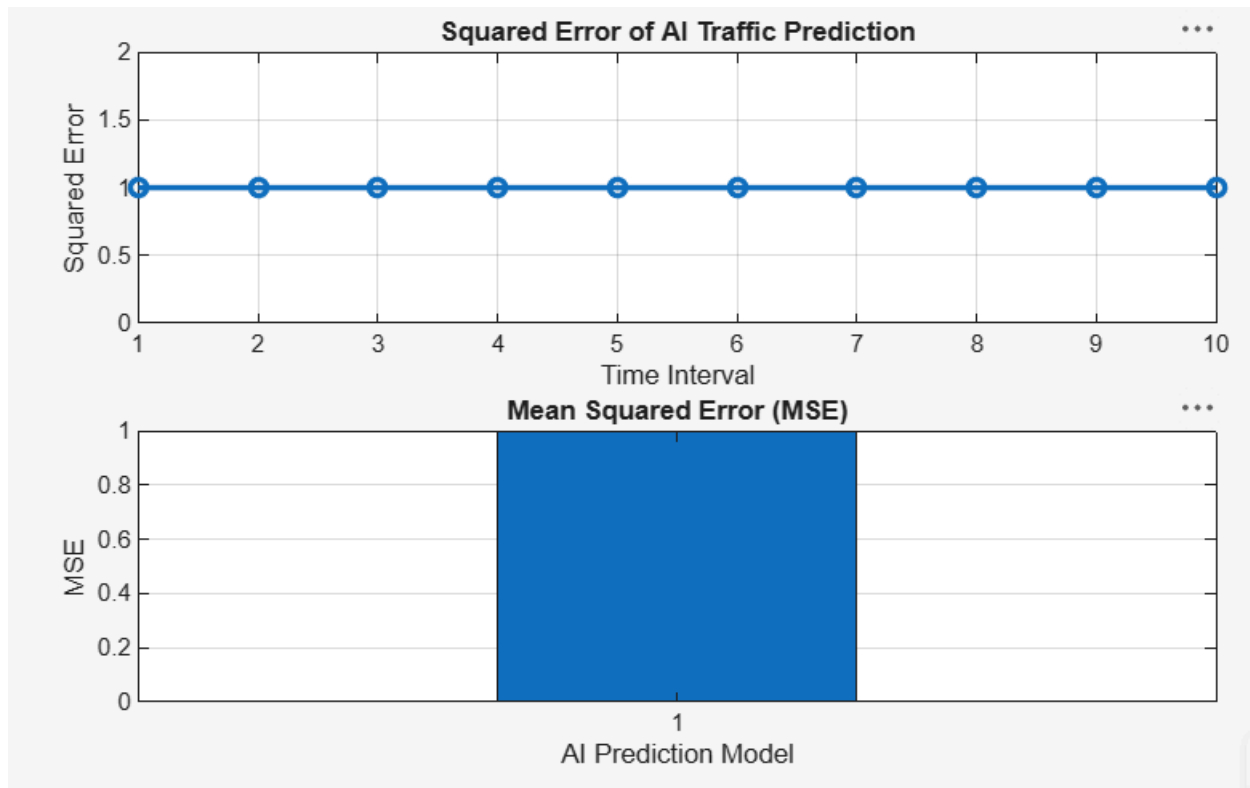
Objective 2 Matlab Code and Output:

```
clc;
clear;
close all;
% Time intervals
t = 1:10;
% Actual and Predicted Traffic (Mbps)
Actual = [50 55 60 62 68 70 75 80 85 90];
Predicted = [49 54 61 63 67 71 74 81 84 89];
% Squared Error
SqError = (Actual - Predicted).^2;
% Mean Squared Error
MSE = mean(SqError);
disp(['Mean Squared Error (MSE) = ', num2str(MSE)])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(t,SqError,'-o','LineWidth',2)
title('Squared Error of AI Traffic Prediction')
xlabel('Time Interval')
ylabel('Squared Error')
grid on
```

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```
% ----- Graph 2 -----  
subplot(2,1,2)  
bar([MSE])  
title('Mean Squared Error (MSE)')  
xlabel('AI Prediction Model')  
ylabel('MSE')  
grid on
```



Objective 3 Matlab Code and Output:

```
clc;  
clear;  
close all;  
% Time intervals  
t = 1:10;  
% Network Throughput (Mbps)  
Throughput = [120 135 150 165 180 195 210 225 240 255];  
% Network Utilization (%)  
Utilization = [60 65 70 74 78 82 86 90 94 98];  
figure  
% ----- Graph 1 -----
```

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```
subplot(2,1,1)
plot(t,Throughput,'-o','LineWidth',2)
title('AI-Based Network Throughput')
xlabel('Time Interval')
ylabel('Throughput (Mbps)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(Throughput,Utilization,'-s','LineWidth',2)
title('Throughput vs Network Utilization')
xlabel('Network Throughput (Mbps)')
ylabel('Network Utilization (%)')
grid on
disp('Time    Throughput(Mbps)    Utilization(%)')
disp([t' Throughput' Utilization'])
```

